

Make Your Own REMOTE-CONTROLLED ROBOT

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To control a robot, the signal is usually sent through a wireless system using radio frequency (RF) and infrared (IR). To send an IR signal, you need a transmitter and a receiver with an encoding and decoding system. The simple circuit presented here can control the movement of a robot using an IR remote that is used for a TV or DVD player, or an equivalent IR remote.

The basic working principle of this project is that the signal sent by the IR remote on pressing Play/Record button on the remote is saved in EEPROM of the microcontroller (MCU) in record mode. It plays back the same code (signal) when you press the same button in play mode. The program for the MCU can save 4-bit code (0 to F hex code) in record mode and output the same 4-bit code when the button is pressed in play mode.

Using the combination of these four bits, two motors installed on either side of the robot (third one being the caster wheel) can be controlled through a motor driver IC. Thus, different codes are used to control different movements of the robot.

Circuit and working

The circuit diagram of the remote-controlled robot is shown in Fig. 1. It is built around 5V voltage regulator 7805 (IC1), ATmega8A MCU (IC2), motor driver IC L293D (IC3), two DC motors (M1 and M2), IR receiver module TSOP1738 (IRRX1), 4-way DIP switch (DIP1), 5mm LEDs (LED1 through LED9) and a few other components.

Once the circuit (or hardware) is ready, burn Remote_Robot.hex code to ATmega8A.

When power switch S1 is switched on, LED2 through LED7 glow for a while and turn off (self-test). Then, either LED6 or LED7 glows, indicating record or play mode, respectively. The mode depends on the setting of jumper J2.

In record mode, J2 is connected to the circuit and in play mode J2 is open. Press the required button on the remote once to record it. Then, permanently set J2 to play mode (that is, remove J2). If you want to change the code to other buttons, connect J2 to record mode.

For initial setup, either disconnect motors M1

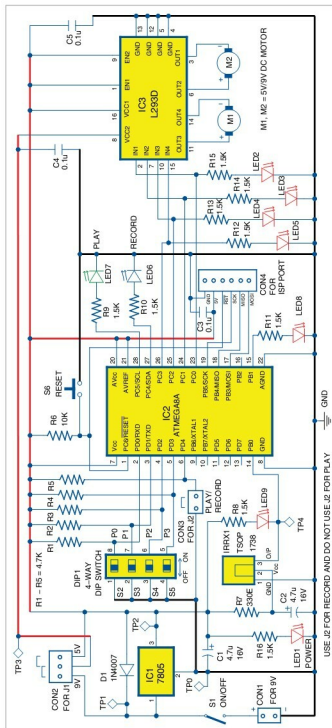


Fig. 1: Circuit diagram of the remote-controlled robot